

3. Why did you choose Next.js instead of React?

Answer

Sir, Next.js is built on top of React and provides several additional features such as file-based routing, server-side rendering (SSR), static site generation (SSG), image optimization, and better SEO. These features improve performance and user experience. That's why we chose Next.js for Vyntra instead of using plain React.

4. What is a REST API?

Answer

REST API stands for Representational State Transfer Application Programming Interface. It is a standard way for the frontend and backend to communicate using HTTP methods such as GET, POST, PUT, and DELETE. The frontend sends a request, the backend processes it, and returns data, usually in JSON format.

5. What is JWT?

Answer

JWT stands for JSON Web Token. It is used for secure user authentication. After a successful login, the server generates a token and sends it to the client. The client stores this token and sends it with future requests. The server verifies the token before allowing access to protected resources.

6. Difference between Authentication and Authorization?

Answer

Authentication means verifying the identity of the user, such as checking the username and password during login.

Authorization means checking what the authenticated user is allowed to access, such as whether the user is an admin or a normal customer.

In simple words:

Authentication = Who are you?

Authorization = What are you allowed to do?

7. What is OAuth?

Answer

OAuth is an authorization framework that allows users to log in using third-party services such as Google or GitHub without sharing their passwords with our application. It makes the login process faster, more secure, and more convenient for users.

8. Why did you use Razorpay?

Answer

Sir, since Vyntra involves booking rental products, online payment was an essential feature. Razorpay provides a secure and reliable payment gateway with support for UPI, cards, net banking, and wallets. It also offers easy integration, which made it a suitable choice for our project.

9. Why Cloudinary?

Answer

Cloudinary is a cloud-based media management service used to store and manage images efficiently. Instead of storing images directly on the server, we uploaded them to Cloudinary and stored only the image URLs in the database. This improves scalability, reduces server storage, and provides automatic image optimization.

10. Explain the Login Flow.

Answer

When a user enters their email and password, the frontend sends a POST request to the login API.

The backend checks whether the user exists and verifies the password.

If the credentials are correct, the backend generates a JWT token and sends it back to the frontend.

The frontend stores this token and includes it in the Authorization header for future requests.

Whenever the user accesses a protected route, the backend verifies the token before returning the requested data.

★ Bonus Questions (TCS Loves These)

Why did you use Tailwind CSS?

Tailwind CSS allows us to build responsive and modern user interfaces quickly using utility classes. It reduces the need to write separate CSS files, improves development speed, and keeps the design consistent.

Why MongoDB?

MongoDB is a NoSQL database that stores data in JSON-like documents. It is flexible, scalable, and well-suited for applications where the data structure may evolve over time.

What is Role-Based Access Control?

Role-Based Access Control (RBAC) restricts access based on user roles. For example, an admin can manage all products and users, while a normal customer can only browse, book, and manage their own rentals. This improves security by ensuring users can access only the features relevant to their role.

If I ask you to build Vyntra again, what would you improve?

I would improve the search and filtering system, add real-time notifications, implement product recommendations, optimize performance further, strengthen security, and improve the UI/UX for an even better user experience.

Perfect bhai. Ab hum **Technology Round** ki preparation karte hain.

Main interviewer ki tarah **sabse expected questions** poochunga aur unke **perfect TCS-level answers** dunga.

MODULE 2: React.js & Next.js (Resume Based)

Q1. What is React?

Answer

React is an open-source JavaScript library developed by Facebook for building user interfaces, especially Single Page Applications (SPAs). It allows us to create reusable UI components, making the application easier to develop, maintain, and scale.

Q2. Why React?

Answer

React provides component-based architecture, Virtual DOM for faster rendering, reusable components, efficient state management, and a large ecosystem. These features help in building modern and responsive web applications efficiently.

Q3. What is Virtual DOM?

Answer

Virtual DOM is a lightweight copy of the real DOM. When the state changes, React first updates the Virtual DOM, compares it with the previous version (called reconciliation), and updates only the changed elements in the real DOM. This improves performance.

👉 **Interviewer Tip:** If they ask "Why is React fast?", this is the answer.

Q4. What is JSX?

Answer

JSX stands for JavaScript XML. It allows us to write HTML-like syntax inside JavaScript. JSX is converted into JavaScript by Babel before execution.

Example:

```
function App() {  
  return <h1>Hello World</h1>;  
}
```

Q5. What are Components?

Answer

Components are reusable and independent pieces of UI. React applications are built by combining multiple components together.

Types:

- Functional Components ✅ (mostly used today)
 - Class Components
-

Q6. Difference between Props and State?

Props	State
Passed from parent	Managed inside component
Read-only	Can change
Used for communication	Used for dynamic data

Q7. What is useState()?

Answer

useState is a React Hook used to store and update data inside a functional component. Whenever the state changes, React automatically re-renders the component.

Example:

```
const [count, setCount] = useState(0);
```

Q8. What is useEffect()?

Answer

useEffect is a React Hook used for side effects such as API calls, fetching data, timers, or event listeners. It runs after the component is rendered.

Example:

```
useEffect(() => {  
  fetchData();  
}, []);
```

Q9. Why Keys are used?

Answer

Keys help React identify which items have changed, been added, or removed in a list. This improves rendering performance and prevents unnecessary re-rendering.

Q10. Controlled vs Uncontrolled Components?

Controlled

React controls input using state.

Uncontrolled

DOM controls input using refs.

NEXT.JS (Very Important)

Q11. What is Next.js?

Answer

Next.js is a React framework that provides additional features like file-based routing, server-side rendering, static site generation, image optimization, and API routes. It helps build faster, SEO-friendly, and production-ready applications.

Q12. Why did you use Next.js instead of React?

Answer

We chose Next.js because it provides built-in routing, better performance, SEO support through server-side rendering, optimized image handling, and a simpler project structure compared to plain React.

Q13. Difference between React and Next.js?

React	Next.js
Library	Framework
Manual Routing	File-based Routing
No SSR by default	Supports SSR
SEO Limited	Better SEO
Needs extra configuration	Built-in features

Q14. What is Server-Side Rendering (SSR)?

Answer

In SSR, the HTML page is generated on the server for every request before being sent to the browser. This improves SEO and allows users to see content faster on the initial page load.

Q15. What is Client-Side Rendering (CSR)?

Answer

In CSR, the browser downloads JavaScript first, and then React renders the UI on the client side. It provides a smooth user experience after the initial load but is generally less SEO-friendly than SSR.

Q16. What is Static Site Generation (SSG)?

Answer

In SSG, pages are generated during the build process and served as static files. This results in very fast page loading and is suitable for content that doesn't change frequently.

Q17. What is File-based Routing?

Answer

In Next.js, routing is automatically created based on the folder and file structure inside the `app` or `pages` directory. There is no need to configure routes manually.

Example:

`app/`

`home/page.js`

`about/page.js`

Automatically creates:

`/home`

`/about`

Q18. What is Hydration? (Advanced but asked sometimes)

Answer

Hydration is the process where React attaches JavaScript functionality to the HTML that was already rendered by the server, making the page interactive.

★ Interview Trick

If interviewer asks:

Which one is better, React or Next.js?

Don't say:

"Next.js is better."

Say:

React is a JavaScript library, while Next.js is a framework built on top of React. If the project requires SEO, routing, and better performance, Next.js is usually a better choice. For smaller applications, React alone may also be sufficient. The choice depends on project requirements.

That answer shows maturity.

🔥 Rapid Fire (30 Seconds)

Q: What is React?

→ JavaScript library for building UI.

Q: What is Next.js?

→ React framework.

Q: What is JSX?

→ HTML-like syntax inside JavaScript.

Q: What is Virtual DOM?

→ Lightweight copy of Real DOM.

Q: useState?

→ Stores component state.

Q: useEffect?

→ Handles side effects.

Q: Props?

→ Read-only data from parent.

Q: State?

→ Component's own mutable data.

Q: SSR?

→ Server renders HTML before sending it.

Q: CSR?

➡ Browser renders UI after loading JavaScript.

Q: SSG?

➡ Pages generated at build time.



Next Module (Most Important)

Ab hum **Node.js + Express.js + REST API + JWT** karenge.

Ye TCS mein bahut frequently poocha jaata hai, aur tumhare Vyntra aur Attendify dono projects se directly linked hai. Agar ye module strong ho gaya, to backend se related basic questions confidently handle kar paoge.